

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 3

Course Code: CSE276

Course Title: ARTIFICIAL INTELLIGENCE
FOUNDATIONS

Course Outcome: CO3

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 3

Unit Title: Intelligent Agents

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the element of the PEAS framework that describes the surroundings in which the agent operates.

- A) Environment
- B) Performance measure
- C) Actuators
- D) Sensors

Q2. Identify the element of the PEAS framework by which an agent acts upon its world.

- A) Sensors
- B) Actuators
- C) Environment
- D) Performance measure

Q3. State the element of the PEAS framework through which an agent perceives its world.

- A) Actuators
- B) Environment
- C) Sensors
- D) Performance measure

Q4. Name the component of a learning agent that selects the actions to be taken.

- A) The critic
- B) The learning element

- C) The problem generator
- D) The performance element

Q5. Identify the component of a learning agent that improves the performance element.

- A) The critic
- B) The learning element
- C) The problem generator
- D) The actuator set

Q6. State the term for a rule of the form: if this condition holds, take this action.

- A) A utility function
- B) A percept sequence
- C) A transition model
- D) A condition-action rule

Q7. Name the agent that looks up its action in a table indexed by the whole percept sequence.

- A) A table-driven agent
- B) A model-based reflex agent
- C) A goal-based agent
- D) A learning agent

Q8. Identify the function an agent uses to compare two outcomes that both satisfy its goal.

- A) A goal test

- B) A condition-action rule
- C) A utility function
- D) A percept function

Q9. State the term for an environment in which the sensors reveal the complete state at every moment.

- A) Partially observable
- B) Deterministic
- C) Fully observable
- D) Episodic

Q10. Name the term for an environment whose rules of operation are not known to the agent in advance.

- A) An unknown environment
- B) A partially observable environment
- C) A stochastic environment
- D) A continuous environment

Q11. Identify what the architecture of an agent consists of.

- A) The set of goals the designer has supplied
- B) The table of percept sequences and actions
- C) The rules held in the agent's knowledge base
- D) The computing device together with its sensors and actuators

Q12. State the term for a multi-agent setting in which all the agents share the same objective.

- A) Competitive
- B) Cooperative
- C) Episodic
- D) Deterministic

Q13. Name the multi-agent setting in which one agent's gain is another agent's loss.

- A) Cooperative
- B) Sequential
- C) Static
- D) Competitive

Q14. Identify the agent type that holds an explicit description of the state it is trying to reach.

- A) A simple reflex agent
- B) A table-driven agent
- C) A goal-based agent
- D) A model-based reflex agent

Q15. State the term for an arrangement in which a person reviews or approves the decisions an agent proposes.

- A) Full autonomy
- B) Human in the loop
- C) Multi-agent coordination
- D) Reflex control

Q16. Identify the four things on which the rationality of an action depends.

- A) The performance measure, prior knowledge, actions and percepts
- B) The sensors, actuators, memory and processor speed
- C) The goal, the utility, the reward and the penalty
- D) The environment, the architecture, the program and the goal

Q17. Explain why rationality is not the same thing as omniscience.

- A) A rational agent always knows every outcome
- B) A rational agent acts best on what it could know
- C) A rational agent is never mistaken about anything
- D) A rational agent must perceive the whole state

Q18. Distinguish a simple reflex agent from a table-driven agent.

- A) A reflex agent uses rules, a table-driven agent stores every case
- B) A table-driven agent uses rules, a reflex agent stores cases
- C) Both of the two store an entry for every percept sequence
- D) Both of the two summarise their behaviour into rules

Q19. Describe what the performance measure of a PEAS specification should be based upon.

- A) The number of actions the agent carries out
- B) The speed at which the agent's program runs
- C) The confidence the agent has in its choices
- D) The state of the environment the agent brings about

Q20. Interpret the following PEAS entry: the area of floor left clean for each unit of energy consumed.

- A) It is an actuator of the agent
- B) It is a sensor of the agent
- C) It is a performance measure
- D) It is a property of the environment

Q21. Predict what a utility-based agent does when no available action fully satisfies its goal.

- A) It selects the action with the highest expected utility
- B) It halts and reports that no solution exists
- C) It selects an action entirely at random

D) It repeats the last action it took before

Q22. Recognize the environment property described: a driving agent whose sensors cannot see round the corner ahead.

- A) Fully observable
- B) Episodic
- C) Partially observable
- D) Static

Q23. Explain why a learning agent needs to take actions it does not currently believe to be best.

- A) Otherwise its sensors gradually stop functioning
- B) Otherwise it never discovers a better course of action
- C) Otherwise its performance measure cannot be computed
- D) Otherwise its actuators become uncalibrated

Q24. Summarize what the agent architecture and the agent program make up between them.

- A) The environment the agent acts in
- B) The performance measure of the agent
- C) The percept sequence of the agent
- D) The agent itself

Q25. Distinguish a cooperative multi-agent system from a competitive one.

- A) In a competitive system the agents share an objective
- B) In a cooperative system there is only one agent
- C) In a cooperative system the agents share an objective
- D) In a competitive system no agent has any objective

Q26. Explain why communication is needed among the agents of a multi-agent system.

- A) Communication removes the need for any sensors
- B) Communication guarantees the agents will agree
- C) Communication reduces the number of agents needed
- D) No agent can observe the whole state on its own

Q27. Interpret the following as an agent type: a thermostat that switches the heating on whenever the temperature falls below a set point.

- A) A simple reflex agent
- B) A goal-based agent
- C) A utility-based agent
- D) A learning agent

Q28. Describe what emergent behaviour means in a multi-agent system.

- A) Behaviour copied from one agent by all the others
- B) Group behaviour that no single agent was programmed with
- C) Behaviour supplied to every agent by the designer
- D) Behaviour that stops as soon as one agent fails

Q29. Explain why a goal-based agent needs a model of how its actions change the world.

- A) It must store every percept it has received
- B) It must compare two outcomes for desirability
- C) It must be able to communicate with other agents
- D) It must judge where a sequence of actions will lead

Q30. Predict the effect of giving every agent in a shared space the same fixed priority rule.

- A) The agents always reach the best joint outcome
- B) Symmetrical situations may leave every agent waiting
- C) The agents no longer need any sensors at all
- D) The agents will always avoid one another safely

Q31. Recognize the collaboration pattern described: the system proposes a shortlist of options and a person makes the final choice.

- A) Full automation with no human involvement
- B) A multi-agent system with competing agents
- C) Human-AI collaboration with a person deciding
- D) A simple reflex agent acting on its percepts

Q32. Explain why a performance measure should be stated in terms of the environment rather than the agent's own behaviour.

- A) Otherwise the agent can satisfy it without doing the task
- B) Otherwise the agent cannot perceive its own state
- C) Otherwise the measure cannot be written down
- D) Otherwise the agent needs a utility function

Q33. Distinguish the sensors of an agent from its actuators.

- A) Sensors perceive the world, actuators change it
- B) Actuators perceive the world, sensors change it
- C) Both of the two perceive the state of the world
- D) Both of the two change the state of the world

Q34. Explain what the internal state of a model-based reflex agent is for.

- A) To store every action the agent has ever taken
- B) To hold the utility of every possible outcome

- C) To record the rules supplied by the designer
- D) To track the part of the world the sensors cannot see

Q35. Interpret an environment in which no other agent is pursuing an objective of its own.

- A) It is a cooperative multi-agent environment
- B) It is a single-agent environment
- C) It is a competitive multi-agent environment
- D) It is a partially observable environment

Q36. Summarize the applications of intelligent agents named on the syllabus.

- A) Image compression and file encryption
- B) Spreadsheet calculation and report printing
- C) Robotic cleaning, warehouse handling and traffic control
- D) Database indexing and network routing tables

Q37. Describe what makes a system an autonomous intelligent system.

- A) Its behaviour is fixed entirely by its designer
- B) It operates without any sensors of its own
- C) Its behaviour is shaped by its own experience
- D) It requires a person to approve every action

Q38. Explain why full automation is not always to be preferred to human-AI collaboration.

- A) A person is always faster than the agent
- B) A person supplies judgement and accountability the agent lacks
- C) A person can perceive more than any sensor
- D) Automation is prohibited in every application

Q39. Predict what a learning agent is able to do that an agent with fixed rules cannot.

- A) Perceive parts of the world without sensors
- B) Act without any performance measure at all
- C) Guarantee that its first action is correct
- D) Improve its behaviour from the outcomes it observes

Q40. Distinguish a goal-based agent from a simple reflex agent by what each considers before acting.

- A) The goal-based agent considers where the action leads
- B) The simple reflex agent considers where it leads
- C) Both of the two consider where the action leads
- D) Neither of the two considers the current percept

----- End of Sample Question Set -----